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The Binomial Formula

Man is a reed, but a reed that thinks.
—BLAISE PASCAL (FRANCE, 1623–1662)

1. INTRODUCTION

This chapter explains how to answer questions like the following.

- A coin is tossed four times. What is the chance of getting exactly one head?
- A die is rolled ten times. What is the chance of getting exactly three aces?
- A box contains one red marble and nine green ones. Five draws are made at random with replacement. What is the chance that exactly two draws will be red?

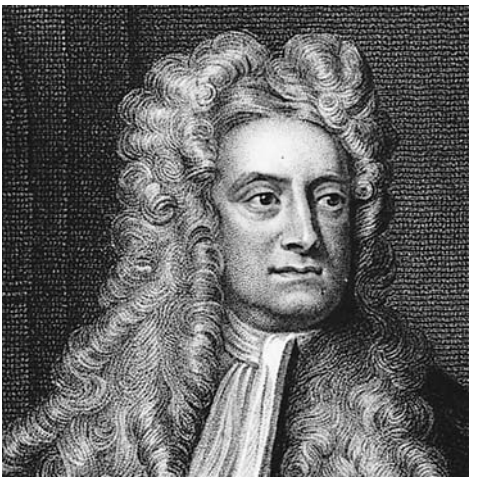
These problems are all similar, and can be solved using the *binomial coefficients*, discovered by Pascal and Newton.[1] The method will be illustrated on the marbles.

The problem is to find the chance of getting two reds (no more and no less) in five draws from the box; so the other three draws must be green. One way this can happen is that the first two draws are red and the final three are green. With R for red and G for green, this possibility can be written

R R G G G

Of course, there are many other ways to get two reds. For example, the second and the fifth draws might be red, while all the rest are green:

G R G G R



Isaac Newton (England, 1642–1727).

From the Warden Collection; engraved by W.T. Fry after a painting by G. Kneller.

To solve the problem, we must find all the possible ways, calculate the chance of each, and then use the addition rule to add up the chances. The first task seems formidable, so we postpone it for a moment and turn to the second one.

The chance of the pattern R R G G G is

$$\frac{1}{10} \times \frac{1}{10} \times \frac{9}{10} \times \frac{9}{10} \times \frac{9}{10} = \left(\frac{1}{10}\right)^2 \left(\frac{9}{10}\right)^3$$

This follows from the multiplication rule: on each draw, the chance of red is 1/10, the chance of green is 9/10.

Similarly, the chance of the pattern G R G G R equals

$$\frac{9}{10} \times \frac{1}{10} \times \frac{9}{10} \times \frac{9}{10} \times \frac{1}{10} = \left(\frac{1}{10}\right)^2 \left(\frac{9}{10}\right)^3$$

The pattern G R G G R has the same chance as the pattern R R G G G. In fact, each pattern with 2 reds and 3 greens has the same chance, $(1/10)^2(9/10)^3$, since the 2 reds will contribute $(1/10)^2$ to the product and the 3 greens will contribute $(9/10)^3$. The sum of the chances of all the patterns, therefore, equals the number of patterns times the common chance.

How many patterns are there? Each pattern is specified by writing down in a row 2 R's and 3 G's, in some order. The number of patterns is given by the *binomial coefficient,*

$$\frac{5 \times 4 \times 3 \times 2 \times 1}{(2 \times 1) \times (3 \times 2 \times 1)} = 10$$

In other words, there are 10 different patterns with 2 R's and 3 G's. So the chance of drawing exactly 2 reds is

$$10 \times \left(\frac{1}{10}\right)^2 \left(\frac{9}{10}\right)^3 \approx 7\%$$

Binomial coefficients look messy. Mathematicians get around this by introducing convenient notation. They use an exclamation mark (!) to indicate the result of multiplying together a number and all the numbers which come before it. For example,

$$\begin{aligned} 1! &= 1 \\ 2! &= 2 \times 1 = 2 \\ 3! &= 3 \times 2 \times 1 = 6 \\ 4! &= 4 \times 3 \times 2 \times 1 = 24 \end{aligned}$$

And so on. The exclamation mark is read "factorial," so that $4! = 24$ is read "four-factorial equals twenty-four." Now the binomial coefficient is easier to read:

$$\frac{5!}{2!\,3!}$$

Remember what the formula represents—the number of different ways of arranging 2 R's and 3 G's in a row.

The 5 in the numerator of the formula is the sum of 2 and 3 in the denominator. Binomial coefficients always take this form. For example, the number of ways to arrange four R's and one G in a row is

$$\frac{5!}{4!\,1!} = 5$$

The patterns are

R R R R G R R R G R R R G R R R G R R R G R R R R

How many ways are there to arrange five R's and zero G's in a row? There is only one way, R R R R R. Applying the formula mechanically gives

$$\frac{5!}{5!\,0!}$$

But we have not yet said what 0! means. It is a convention of mathematics that $0! = 1$. With this convention, the binomial coefficient does equal 1.

Binomial coefficients and factorials get very large very quickly. For instance, the number of ways to arrange 10 R's and 10 G's in a row is given by the binomial coefficient

$$\frac{20!}{10!\,10!} = 184{,}756$$

However, there was a lot of cancellation going on: $10! = 3{,}628{,}800$; and $20! \approx 2 \times 10^{18}$, or 2 followed by 18 zeros. (A trillion is 1 followed by 12 zeros.)

Exercise Set A

1. Find the number of different ways of arranging one R and three G's in a row. Write out all the patterns.

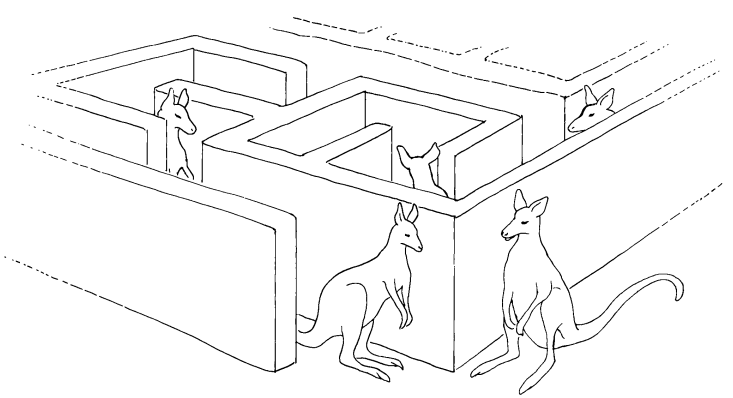
2. Find the number of different ways of arranging two R's and two G's in a row. Write out all the patterns.

3. A box contains one red ball and five green ones. Four draws are made at random with replacement from the box. Find the chance that—
   (a) a red ball is never drawn
   (b) a red ball appears exactly once
   (c) a red ball appears exactly twice
   (d) a red ball appears exactly three times
   (e) a red ball appears on all the draws
   (f) a red ball appears at least twice

4. A die is rolled four times. Find the chance that—
   (a) an ace (one dot) never appears
   (b) an ace appears exactly once
   (c) an ace appears exactly twice

5. A coin is tossed 10 times. Find the chance of getting exactly 5 heads. Find the chance of obtaining between 4 and 6 heads inclusive.

6. It is claimed that a vitamin supplement helps kangaroos learn to run a special maze with high walls. To test whether this is true, 20 kangaroos are divided up into 10 pairs. In each pair, one kangaroo is selected at random to receive the vitamin supplement; the other is fed a normal diet. The kangaroos are then timed as they learn to run the maze. In 7 of the 10 pairs, the treated kangaroo learns to run the maze more quickly than its untreated partner. If in fact the vitamin supplement has



"Did you bring the vitamins ?"

no effect, so that each animal of the pair is equally likely to be the quicker, what is the probability that 7 or more of the treated animals would learn the maze more quickly than their untreated partners, just by chance?

The answers to these exercises are on pp. A70–71.

2. THE BINOMIAL FORMULA

The reasoning of section 1 is summarized in the *binomial formula*. Suppose a chance process is carried out as a sequence of trials. An example would be rolling a die 10 times, where each roll counts as a trial. There is an event of interest which may or may not occur at each trial: the die may or may not land ace. The problem is to calculate the chance that the event will occur a specified number of times.

> The chance that an event will occur exactly k times out of n is given by the binomial formula
>
> $$\frac{n!}{k!\,(n-k)!}\,p^k(1-p)^{n-k}$$
>
> In this formula, n is the number of trials, k is the number of times the event is to occur, and p is the probability that the event will occur on any particular trial. The assumptions:
>
> - The value of n must be fixed in advance.
> - p must be the same from trial to trial.
> - The trials must be independent.

The formula starts with the binomial coefficient (p. 256),

$$\frac{n!}{k!\,(n-k)!}$$

Remember, this is the number of ways to arrange n objects in a row, when k are alike of one kind and $n-k$ are alike of another (for instance, red and green marbles).

Example 1. A die is rolled 10 times. What is the chance of getting exactly 2 aces?

Solution. The number of trials is fixed in advance. It is 10. So $n = 10$. The event of interest is rolling an ace. The probability of rolling an ace is the same from trial to trial. It is 1/6. So $p = 1/6$. The trials are independent. The binomial formula can be used, and the answer is

$$\frac{10!}{2!\,8!}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^8 \approx 29\%$$

Example 2. A die is rolled until it first lands six. If this can be done using the binomial formula, find the chance of getting 2 aces. If not, why not?

Solution. The number of trials is not fixed in advance. It could be 1, if the die lands six right away. Or it could be 2, if the die lands five then six. Or it could be 3. And so forth. The binomial formula does not apply.

Example 3. Ten draws are made at random with replacement from the box $\boxed{1}\ \boxed{1}\ \boxed{2}\ \boxed{3}\ \boxed{4}\ \boxed{5}$. However, just before the last draw is made, whatever else has gone on, the ticket $\boxed{5}$ is removed from the box. True or false: the chance of drawing exactly two $\boxed{1}$'s is

$$\frac{10!}{2!\,8!}\left(\frac{2}{6}\right)^2\left(\frac{4}{6}\right)^8$$

Solution. In this example, n is fixed in advance and the trials are independent. However, p changes at the last trial from $2/6$ to $2/5$. So the binomial formula does not apply, and the statement is false.

Example 4. Four draws are made at random without replacement from the box in example 3. True or false: the chance of drawing exactly two $\boxed{1}$'s is

$$\frac{4!}{2!\,2!}\left(\frac{2}{6}\right)^2\left(\frac{4}{6}\right)^2$$

Solution. The trials are dependent, so the binomial formula does not apply.

Technical notes. (i) To work out the chance in example 4, take a pattern with exactly two 1's, like 1 1 N N, where N means "not 1." The chance of getting 1 1 N N equals

$$\frac{2}{6}\times\frac{1}{5}\times\frac{4}{4}\times\frac{3}{3}=\frac{1}{15}$$

Surprisingly, the chance is the same for all such patterns. How many patterns have exactly two 1's? The answer is

$$\frac{4!}{2!\,2!}=6$$

So the chance of getting exactly two 1's is

$$6\times\frac{1}{15}=\frac{2}{5}$$

(ii) Mathematicians usually write $\binom{n}{k}$ for the binomial coefficient:

$$\binom{n}{k}=\frac{n!}{k!\,(n-k)!}$$

They read $\binom{n}{k}$ as "n choose k," the idea being that the formula gives the number of ways to choose k things out of n. Older books write the binomial coefficient as ${}_nC_k$ or ${}^nC_k$, the "number of combinations of n things taken k at a time."

3. REVIEW EXERCISES

Review exercises may cover material from previous chapters.

1. A die will be rolled 6 times. What is the chance of obtaining exactly 1 ace?

2. A die will be rolled 10 times. The chance it never lands six can be found by one of the following calculations. Which one, and why?

$$\text{(i) } \left(\frac{1}{6}\right)^{10} \qquad \text{(ii) } 1-\left(\frac{1}{6}\right)^{10} \qquad \text{(iii) } \left(\frac{5}{6}\right)^{10} \qquad \text{(iv) } 1-\left(\frac{5}{6}\right)^{10}$$

3. Of families with 4 children, what proportion have more girls than boys? You may assume that the sex of a child is determined as if by drawing at random with replacement from[2]

$$\boxed{\boxed{\text{M}}\boxed{\text{F}}} \qquad \text{M} = \text{male}, \ \text{F} = \text{female}$$

4. A box contains 8 red marbles and 3 green ones. Six draws are made at random without replacement. True or false: the chance that the 3 green marbles are drawn equals

$$\frac{6!}{3!\,3!}\left(\frac{8}{11}\right)^3\left(\frac{3}{11}\right)^3$$

Explain briefly.

5. There are 8 people in a club.[3] One person makes up a list of all the possible committees with 2 members. Another person makes up a list of all the possible committees with 5 members. True or false: the second list is longer than the first. Explain briefly.

6. There are 8 people in a club. One person makes up a list of all the possible committees with 2 members. Another person makes up a list of all the possible committees with 6 members. True or false: the second list is longer than the first. Explain briefly.

7. A box contains one red marble and nine green ones. Five draws are made at random with replacement. The chance that exactly two draws will be red is

$$10 \times \left(\frac{1}{10}\right)^2\left(\frac{9}{10}\right)^3$$

Is the addition rule used in deriving this formula? Answer yes or no, and explain carefully.

8. A coin will be tossed 10 times. Find the chance that there will be exactly 2 heads among the first 5 tosses, and exactly 4 heads among the last 5 tosses.

9. For each question (a–e) below, choose one of the answers (i–viii); explain your choice.

   Questions

   A deck of cards is shuffled. What is the chance that—

   (a) the top card is the king of spades and the bottom card is the queen of spades?
   (b) the top card is the king of spades and the bottom card is the king of spades?
   (c) the top card is the king of spades or the bottom card is the king of spades?
   (d) the top card is the king of spades or the bottom card is the queen of spades?
   (e) of the top and bottom cards, one is the king of spades and the other is the queen of spades?

   Answers

   (i) $1/52 \times 1/51$
   (ii) $1/52 + 1/51$
   (iii) $1/52 \times 1/52$
   (iv) $1/52 + 1/52$
   (v) $1 - (1/52 \times 1/51)$
   (vi) $1 - (1/52 \times 1/52)$
   (vii) $2/52 \times 1/51$
   (viii) None of the above

10. A box contains 3 red tickets and 2 green ones. Five draws will be made at random. You win \$1 if 3 of the draws are red and 2 are green. Would you prefer the draws to be made with or without replacement? Why?

11. It is now generally accepted that cigarette smoking causes heart disease, lung cancer, and many other diseases. However, in the 1950s, this idea was controversial. There was a strong association between smoking and ill-health, but association is not causation. R. A. Fisher advanced the "constitutional hypothesis:" there is some genetic factor that disposes you both to smoke and to die.

    To refute Fisher's idea, the epidemiologists used twin studies. They identified sets of smoking-discordant monozygotic twin pairs. ("Monozygotic" twins come from one egg and have identical genetic makeup; "smoking-discordant" means that one twin smokes, the other doesn't.) Now there is a race. Which twin dies first, the smoker or the non-smoker? Data from a Finnish twin study are shown at the top of the next page.[4]

Data from the Finnish twin study

	Smokers	*Non-smokers*
All causes	17	5
Coronary heart disease	9	0
Lung cancer	2	0

According to the first line of the table, there were 22 smoking-discordant monozygotic twin pairs where at least one twin of the pair died. In 17 cases, the smoker died first; in 5 cases, the non-smoker died first. According to the second line, there were 9 pairs where at least one twin died of coronary heart disease; in all 9 cases, the smoker died first. According to the last line, there were 2 pairs where at least one twin died of lung cancer, and in both pairs the smoker won the race to death. (Lung cancer is a rare disease, even among smokers.)

For parts (a–c), suppose that each twin in the pair is equally likely to die first, so the number of pairs in which the smoker dies first is like the number of heads in coin-tossing.

(a) On this basis, what is the chance of having 17 or more pairs out of 22 where the smoker dies first?
(b) Repeat the test in part (a), for the 9 deaths from coronary heart disease.
(c) Repeat the test in part (a), for the 2 deaths from lung cancer.
(d) Can the difference between the death rates for smoking and non-smoking twins be explained by
    (i) chance?
    (ii) genetics?
    (iii) health effects of smoking?

4. SPECIAL REVIEW EXERCISES

These exercises cover all of parts I–IV.

1. In the U.S. in 1990, 20,273 people were murdered, compared to 16,848 in 1970—nearly a 20% increase. "These figures show that the U.S. became a more violent society over the period 1970–1990." True or false, and explain briefly.[5]

2. A leading cause of death in the U.S. is coronary artery disease—a breakdown of the main arteries to the heart. The disease can be treated with coronary bypass surgery (section 3 of chapter 1). In one of the first trials, Dr. Daniel Ullyot and associates performed coronary bypass surgery on a test group of patients: 98% survived 3 years or more. The conventional treatment used drugs and special diets to reduce blood pressure and eliminate fatty deposits in the arteries. According to previous studies, only 68% of the patients getting the conventional treatment survived 3 years or more. [Exercise continues. . . .]

A newspaper article described Ullyot's results as "spectacular," because the survival rate among Ullyot's patients was much higher than the survival rate in previous studies.[6]

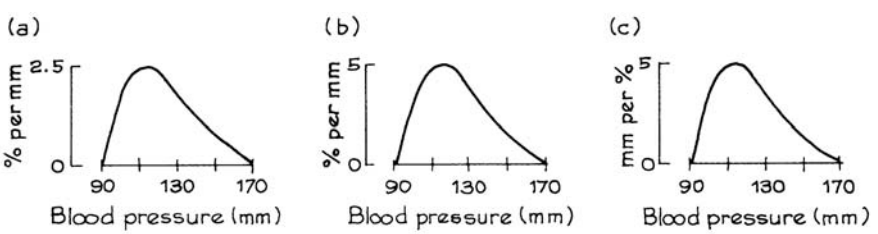
(a) Did Ullyot's study have contemporaneous controls? If not, what patients were used as the comparison group?
(b) Was the newspaper article's enthusiasm justified by the study? Discuss briefly.

3. Susan Bouman was denied a promotion to sergeant in the Los Angeles County Sheriff's Department after she took a competitive exam for the position. She filed suit in federal court in April 1980, claiming that the exam was discriminatory.[7] Data for 1975 and 1977 are shown below. In 1975, the pass rate for women was 10/79 = 12.7%; for men, it was 250/1,312 = 19.1%. The "selection ratio" was 12.7/19.1 = 66.5%: in other words, the women's pass rate was only 66.5% of the men's pass rate. For 1977, the selection ratio was 67.1%. Selection ratios below 80% are generally regarded by the Equal Opportunity Employment Commission as showing "adverse impact" on a "protected group."

   Results can also be analyzed by "pooling" data for the two years—just adding up the numbers. For the two years combined, there were 102 + 79 = 181 women applicants of whom 10 + 18 = 28 passed, and so forth. True or false, and explain: "The selection ratio was 66.5% in 1975 and 67.1% in 1977; therefore, the selection ratio for the pooled data must be between 66.5% and 67.1%."

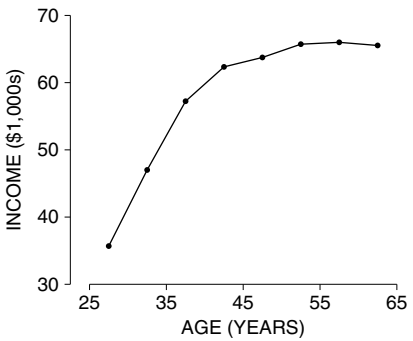
	Women	*Men*
1975		
Applicants	79	1,312
Passed the exam	10	250
1977		
Applicants	102	1,259
Passed the exam	18	331

4. Three people have tried to sketch the histogram for blood pressures of the subjects in a certain study, using the density scale. Only one is right. Which one, and why?



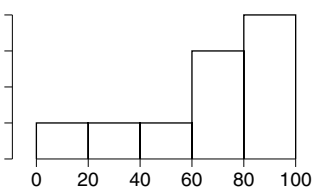
5. A study is made of the age at entrance of college freshmen.[8] Is the SD about 1 month, 1 year, or 5 years. Why?

6. A study is based on a representative sample of men age 25–64 in 2005, who were working full time. The figure below plots average income for each age group.[9] True or false, and explain: the data show that on average, if a man keeps working, his income will increase until age 50 or so, then stabilize. If false, how do you account for the pattern in the data?



Source: March 2005 Current Population Survey; CD-ROM supplied by the Bureau of the Census.

7. True or false, and explain: for the histogram below, the 60th percentile is equal to twice the 30th percentile. (You may assume the distribution is uniform on each class interval.)



8. True or false, and explain. (You don't need to compute the average or the SD of the lists.)

   (a) The following two lists are the same, when converted to standard units:

(i)	1	3	4	7	9	9	9	21	32
(ii)	3	7	9	15	19	19	19	43	65

   (b) The following two lists are the same, when converted to standard units:

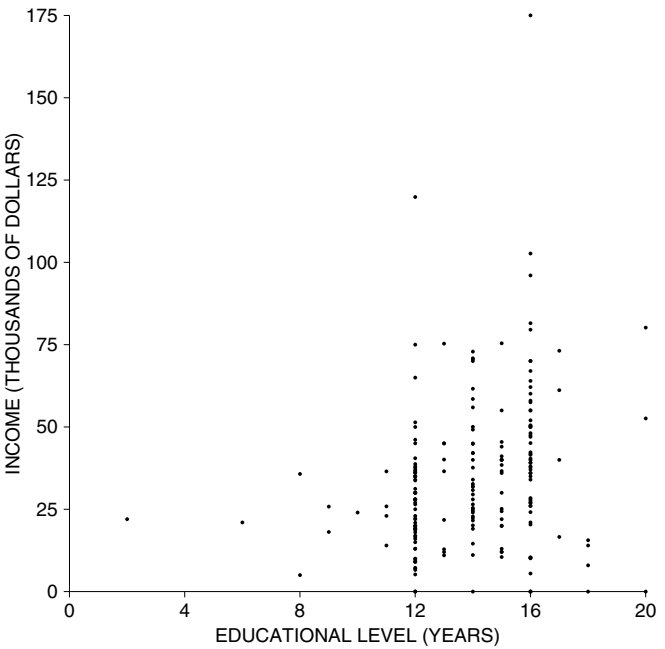
(i)	1	3	4	7	9	9	9	21	32
(ii)	−1	−5	−7	−13	−17	−17	−17	−41	−63

9. In a large class, the average score on the final was 50 out of 100, and the SD was 20. The scores followed the normal curve.

   (a) Two brothers took the final. One placed at the 70th percentile and the other was at the 80th percentile. How many points separated them?

   (b) Two sisters took the final. One placed at the 80th percentile and the other was at the 90th percentile. How many points separated them?

10. The figure below is a scatter plot of income against education (years of schooling completed) for a representative sample of men age 25–34 in Kansas. Or is something wrong? Explain briefly.



11. (a) Find the correlation coefficient for the data set in table (i) below.
    (b) If possible, fill in the blanks in table (ii) below so the correlation coefficient is 1. If this is not possible, explain why not.

(i)

x	y
4	7
5	0
7	9
8	9
8	13
10	16

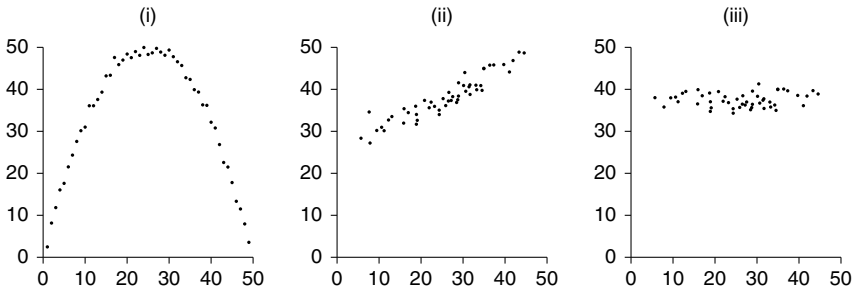
(ii)

x	y
__	7
5	__
7	9
8	9
8	13
10	__

12. In a study of Danish draftees, T. W. Teasdale and associates found a positive correlation between near-sightedness and intelligence.[10] True or false, and explain:

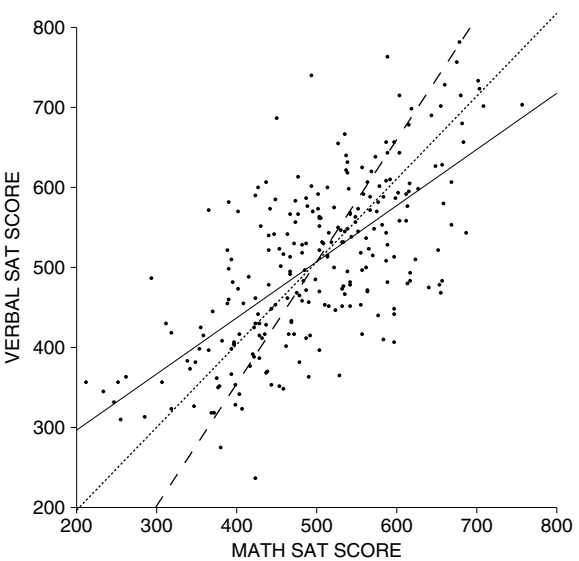
    (a) Draftees who were more near-sighted were also more intelligent, on average.
    (b) Draftees who were more intelligent were also more near-sighted, on average.
    (c) The data show that near-sightedness causes intelligence.
    (d) The data show that intelligence causes near-sightedness.

13. For each diagram below, say whether r is nearly -1, 0, or 1. Explain briefly.



14. The figure below shows a scatter diagram for test scores. Verbal SAT is plotted on the vertical axis and Math SAT on the horizontal. Three lines are drawn across the diagram. Match the line with the description (one description will be left over). Explain briefly.

    (i) estimated average score on V-SAT for given score on M-SAT
    (ii) estimated average score on M-SAT for given score on V-SAT
    (iii) nearly equal percentile ranks on both tests
    (iv) total score on the two tests is about 1,100



15. At a certain law school, first-year scores average 65 and the SD is 12. The correlation between LSAT scores and first-year scores is 0.55. The scatter diagram is football-shaped. The dean's office uses regression to predict first-year scores from LSAT scores. About what percent of the students do better than predicted, by 10 points or more? Explain your answer. If you need more information, say what you need and why.

16. The great prime ministers of France generally served under kings who were less talented. Similarly, the great kings typically had prime ministers who were not as great. Is this a fact of French history, or of statistics? Discuss briefly.

17. In a large class, the midterm had an average of 50 points with an SD of 22. The final scores averaged out to 60 with an SD of 20. The correlation between midterm and final scores was 0.60. The scatter diagram was football-shaped. Of the students who scored around 50 on the midterm, about what percentage were in the top 25% of the class on the final?

18. One ticket is drawn at random from each of the two boxes below:

(A) $\boxed{1}\ \boxed{2}\ \boxed{3}\ \boxed{4}\ \boxed{5}$ (B) $\boxed{1}\ \boxed{2}\ \boxed{3}\ \boxed{4}\ \boxed{5}\ \boxed{6}$

Find the chance that—

(a) One of the numbers is 2 and the other is 5.
(b) The sum of the numbers is 7.
(c) One number is bigger than twice the other.

19. There are 52 cards in a deck, and 13 of them are hearts.

(a) Four cards are dealt, one at a time, off the top of a well-shuffled deck. What is the chance that a heart turns up on the fourth card, but not before? Explain briefly.
(b) A deck of cards is shuffled. You have to deal one card at a time until a heart turns up. You have dealt 3 cards, and still have not seen a heart. What is the chance of getting a heart on the 4th card? Explain briefly.

20. A coin is tossed 10 times. Find the chance of getting 7 heads and 3 tails.

5. SUMMARY AND OVERVIEW

1. The *binomial coefficient* $\dfrac{n!}{k!\,(n-k)!}$ gives the number of ways to arrange n objects in a row, when k are alike of one kind and $n-k$ are alike of another kind (for instance, red and blue marbles).

2. The chance that an event will occur exactly k times out of n is given by the *binomial formula*

$$\frac{n!}{k!\,(n-k)!}\,p^k(1-p)^{n-k}$$

In this formula, n is the number of trials, k is the number of times the event is to occur, and p is the probability that the event will occur on any particular trial. The assumptions:

- The value of n must be fixed in advance.
- p must be the same from trial to trial.
- The trials must be independent.

3. This part of the book defined conditional probabilities, independence, and the multiplication rule. The addition rule was introduced for mutually exclusive events.

4. The binomial formula is an application of the multiplication rule combined with the addition rule.

5. Independence is the basis for the statistical theory to be developed in part V, and the crucial assumption behind many of the procedures to be discussed in parts VI–VIII.